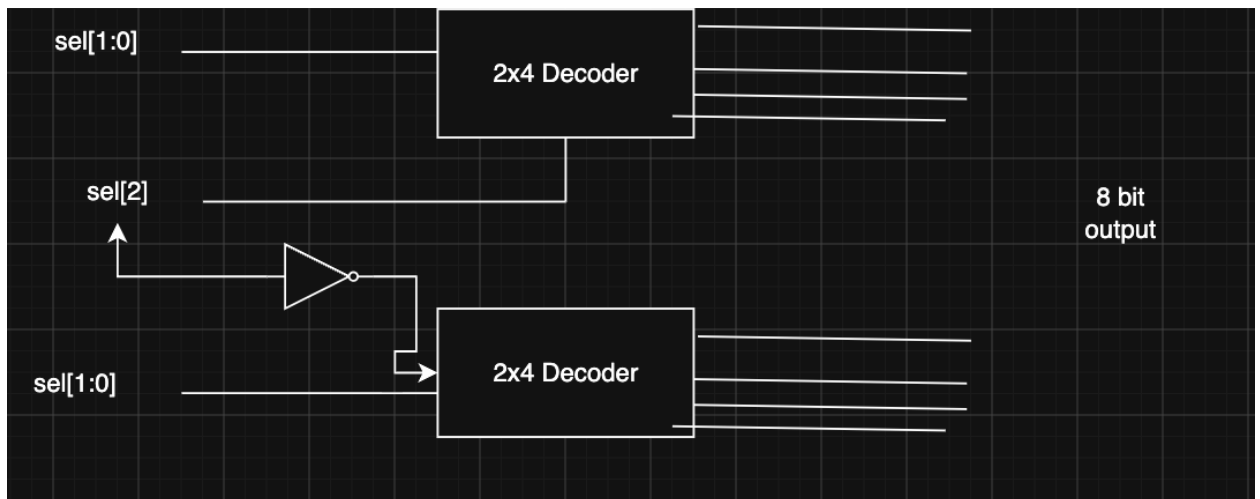


MEDS Module 4 Section 7

Name: Umer Faisal

TASK 6: 3 to 8 decoder from 2 to 4 decoders

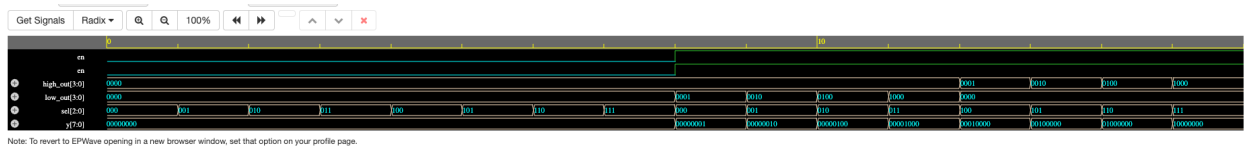
1. EDA Playground Link: https://edaplayground.com/x/Q_iu
2. Schematic Diagram:



3. Terminal Output Screenshot:

```
Log Share
CPU time: .573 seconds to compile + .448 seconds to elab + .563 seconds to link
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 23 17:46 2026
[PASS] Enable=0 sel=0
[PASS] Enable=0 sel=1
[PASS] Enable=0 sel=2
[PASS] Enable=0 sel=3
[PASS] Enable=0 sel=4
[PASS] Enable=0 sel=5
[PASS] Enable=0 sel=6
[PASS] Enable=0 sel=7
[PASS] Enable=1 sel=0
[PASS] Enable=1 sel=1
[PASS] Enable=1 sel=2
[PASS] Enable=1 sel=3
[PASS] Enable=1 sel=4
[PASS] Enable=1 sel=5
[PASS] Enable=1 sel=6
[PASS] Enable=1 sel=7
All tests completed.
$finish called from file "design.sv", line 56.
```

4. Waveform Screenshot:



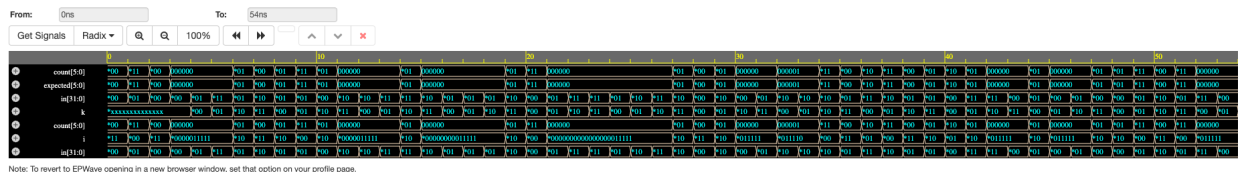
5. Explanation: 2 select bits are the select input for the 2 decoders. And the third select bit is the enable signal for the decoders. So the two 4 bit outputs are concatenated to form the 8 bit output.

TASK 7: 32 bit leading count zero

1. EDA Playground Link: <https://edaplayground.com/x/VxJ9>
2. Terminal Output Screenshot:

```
CPU time: .641 seconds to compile + .563 seconds to elab + .609 seconds to link
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 23 18:13 2026
[PASS] in=00000000 count=32
[PASS] in=00000001 count=31
[PASS] in=00008000 count=16
[PASS] in=80000000 count=0
[PASS] in=c0895e81 count=0
[PASS] in=b1f05663 count=0
[PASS] in=46df998d count=1
[PASS] in=89375212 count=0
[PASS] in=06d7cd0d count=5
[PASS] in=1e8dcd3d count=3
[PASS] in=462df78c count=1
[PASS] in=e33724c6 count=0
[PASS] in=d513d2aa count=0
[PASS] in=bbd27277 count=0
[PASS] in=47ecdb8f count=1
[PASS] in=e77696ce count=0
[PASS] in=e2ca4ec5 count=0
[PASS] in=de8e28bd count=0
[PASS] in=b2a72665 count=0
[PASS] in=0573870a count=5
[PASS] in=10642120 count=3
[PASS] in=cecccc9d count=0
[PASS] in=8983b813 count=0
[PASS] in=a9a7d653 count=0
[PASS] in=eea62ad5 count=0
[PASS] in=d7563eae count=0
[PASS] in=e7c572cf count=0
[PASS] in=0509650a count=5
[PASS] in=9e314c3c count=0
[PASS] in=452e618a count=1
[PASS] in=ec4b34d8 count=0
[PASS] in=c48a1289 count=0
[PASS] in=5b0265b6 count=1
```

3. Waveform Screenshot:



4. Explanation:

Design Explanation

The 32-bit Leading Zero Counter (LZC) counts the number of consecutive zeros starting from the most significant bit (MSB). The implemented design uses a **for-loop** inside an `always_comb` block that scans the input from bit 31 down to bit 0. As soon as the first 1 is found, the number of leading zeros is calculated as $31 - i$. If no 1 is found, the input is all zeros and the output is 32.

A faster alternative is a **tree-structured LZC**, which divides the input into smaller groups (2-bit, 4-bit, 8-bit, etc.) and processes them in parallel. Each stage combines the results from the previous stage until the final leading zero count is obtained. This reduces the number of logic levels in the critical path.

Critical Path Comparison

The linear implementation has a critical path that grows with the input size because it may need to check every bit before finding the first 1. For a 32-bit input, the worst-case path is approximately **32 logic levels**. In contrast, the tree-structured implementation combines results in parallel, requiring only about $\log_2(32) = 5$ **logic levels**. Therefore, the tree version is significantly faster for wider inputs because it reduces propagation delay.